CURRICULUM VITAE

Nimish Sharma

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EDUCATION

Five-year integrated BS-MS in Physics with Minor in Computer Science August 2021 – May 2026 (Expected)
Indian Institute of Science Education and Research Kolkata, India
Thesis Topic: Enhancement of Chiroptical signals for nanoparticles via structured light beams
Details: Chirality plays a fundamental role in numerous biological processes and technological applications including drug development, enantioselective catalysis, and optical devices. However, natural chiral responses are typically weak and challenging to exploit effectively. This research focuses on enhancing chiroptical signals in nanoparticles embedded in a chiral polymer through interactions with structured light. We develop a custom Mie scattering theory for tightly focused beams in chiral host media to support experimental investigations of chiral nanoparticle systems. Our approach combines dark-field polarization microscopy with computational modelling to analyze Mueller matrices in chiral polymers. By employing tight focusing configurations and structured light beams, we report significant enhancement of higher-order quadrupolar modes, enabling amplified differential chiroptical responses. This work provides new pathways for enhancing and exploiting chiral light-matter interactions in nanophotonic systems.
Supervisor: Prof. N. Ghosh (Bio-NaP Lab)

PROFESSIONAL EXPERIENCE

Teaching Assistant - Modern Physics and Optics Lab August 2025 – Dec 2025 (Expected)
Indian Institute of Science Education and Research Kolkata, India

- Assisted in conducting laboratory sessions for modern physics and optics experiments
- Guided students in experimental techniques, data analysis, and report writing
- Provided instructional support for optical instrumentation and measurement techniques

Summer Research Fellow May 2024 – July 2024
SVNIT Surat, India (IAS Summer Research Fellowship Program)
Email: hp@phy.svnit.ac.in

- Conducted first-principles investigations on Heusler alloys using Density Functional Theory
- Performed structural optimizations and calculated magnetic, elastic, and electronic properties
- Utilized Quantum ESPRESSO and WIEN2k software packages for materials simulation
- Replicated and verified results from published research on Ti2VGe Heusler compounds
- Compiled comprehensive report on DFT theory, software implementation, and research findings

RESEARCH EXPERIENCE

Masters Student January 2025 – Present
Bio-NaP Lab, IISER Kolkata
Email: nghosh@iiserkol.ac.in

Light-Matter Interaction of Nanoparticles in Chiral Host Media

- Developed setup for enhancing optical properties of chiral nanoparticles via tight focusing of light
- Automated Mueller matrix measurement procedures through MATLAB scripting
- Building custom Mie scattering theory for tightly focused light beams in chiral host media as part of MS thesis
- Implementing computational models for scattering matrices, Mueller matrices, and optical properties (circular dichroism, optical rotation) using Python and MATLAB
- Manuscript for this work is under preparation

Darkfield Microscopy and Plasmonics

- Utilized darkfield microscopy to measure k-domain Mueller matrices for waveguide plasmonic crystals

- Integrated MATLAB simulations with experimental data to confirm far-field Mueller matrix measurements
- Investigated spin-orbit interaction effects including optical spin Hall effect
- Conducted experimental research on light-matter interactions in various phase gradient metasurfaces
- Analyzed spectral and momentum domain Mueller matrices for advanced optical characterization.
- Manuscript for this work is under preparation.

Quantum Computing Research Intern May 2023 – July 2023
IISER Bhopal, India
- Studied quantum computing fundamentals including quantum gates, entanglement, and algorithms
- Developed quantum deep learning network for binary classification of Iris dataset
- Tested quantum circuits on IBMQManila and compared performance with classical ML models
- Implemented and analyzed quantum teleportation protocols

PUBLICATIONS

Research Manuscripts in Preparation:
Sharma, N., Ghosh, N. (2025) "Enhancement of Chiroptical signals of nanoparticles via Tight Focusing of Light". Manuscript will be in preparation.
Type: Research Article | **Refereed:** Not yet submitted
Authorship: Developed experimental setup for enhancing optical properties of chiral nanoparticles via tight focusing of light. Automated Mueller matrix measurement procedures through MATLAB scripting. Building custom Mie scattering theory for tightly focused light beams in chiral host media as part of MS thesis. Implementing computational models for scattering matrices, Mueller matrices, and optical properties (circular dichroism, optical rotation) using Python and MATLAB.

Sharma, N., Ghosh, N. (2025) "Phase Gradient Mueller Matrix Analysis of Waveguide Plasmonic Crystals". Manuscript will be in preparation.
Type: Research Article | **Refereed:** Not yet submitted
Authorship: Utilized darkfield microscopy to measure k-domain Mueller matrices for waveguide plasmonic crystals. Integrated MATLAB simulations with experimental data to confirm far-field Mueller matrix measurements. Investigated spin-orbit interaction effects including optical spin Hall effect. Conducted experimental research on light-matter interactions in various phase gradient metasurfaces. Analyzed spectral and momentum domain Mueller matrices for advanced optical characterization.

Term Papers and Technical Reports:
Sharma, N. (2025) "Coherent Population Trapping: Theoretical Foundations and Applications". Term Paper. Indian Institute of Science Education and Research Kolkata.
Type of Publication: Term Paper
Refereed Publication: No
Authorship Statement: Comprehensive review of coherent population trapping phenomena, covering theoretical background, experimental implementations, and applications in atomic clocks, magnetometry, and quantum information processing. Includes mathematical derivations of dark states and coherence effects in three-level atomic systems.

Sharma N., Saxena A., Poluri G. (2025) "Dark Matter Profiling through Rotation Curve Analysis of Spiral Galaxies". Term Paper. Indian Institute of Science Education and Research Kolkata.
Type of Publication: Term Paper
Refereed Publication: No
Authorship Statement: Modeled rotation curves of eight spiral galaxies using THINGS HI 21-cm data to study stellar, gas, and dark matter contributions. Compared NFW, pseudo-isothermal, and cored-NFW halo models through rotation curve decomposition and fitting. Demonstrated dark matter dominance at large radii and investigated baryonic feedback effects on inner halo structure.

Sharma, N. (2024) "Metasurfaces and Flat Optics: Spin-Orbit Interactions and Applications". Term Paper. Indian Institute of Science Education and Research Kolkata.
Type of Publication: Term Paper
Refereed Publication: No

Authorship Statement: Investigated optical spin Hall effect and spin-orbit interactions in metasurfaces. Mathematically derived gradient effects of refractive indices in anisotropic media relevant to metasurface fabrication. Comprehensive review of current research in flat optics and metamaterial applications.

Sharma N., Saxena A., Poluri G. (2024) "Casimir Effect in 1D Dirac Systems with Finite Potential Barriers". Term Paper. Indian Institute of Science Education and Research Kolkata.
Type of Publication: Term Paper
Refereed Publication: No
Authorship Statement: Studied quantum Casimir forces between finite potential barriers in 1D systems using massless Dirac fermions. Applied Hellmann-Feynman theorem and matrix methods to calculate forces without mathematical divergences. Demonstrated force switching from attractive to repulsive based on barrier spin orientation, with implications for nano-device design.

Sharma, N. (2024) "First Principle Investigations on Heusler Alloys: DFT Analysis of Structural and Electronic Properties". Research Report. SVNIT Surat (IAS SRFP).
Type of Publication: Research Report
Refereed Publication: No
Authorship Statement: Comprehensive report on density functional theory applications to Heusler alloys. Documented structural optimizations, magnetic properties, elastic properties, dynamical stability, and electronic structure calculations using Quantum ESPRESSO and WIEN2k. Included verification studies replicating published results on Ti2VGe compounds.

Sharma, N. (2023) "Quantum Computing and Machine Learning: Implementation of Quantum Classifiers". Internship Report. IISER Bhopal.
Type of Publication: Internship Report
Refereed Publication: No
Authorship Statement: Developed and implemented quantum deep learning network for Iris dataset classification. Conducted comparative analysis with classical machine learning models including logistic regression, K-nearest neighbors, and decision tree classifiers. Tested quantum circuits on IBMQManila hardware and analyzed performance metrics.

NON-UNIVERSITY PROFESSIONAL ACTIVITIES

Event Organizer January 2025 – Present
OPTICA Student Chapter
IISER Kolkata, India
Nature and extent: Serve as Event Organizer for OPTICA (formerly OSA) Student Chapter at IISER Kolkata. Organize technical talks, seminars, and science outreach programs for local school children. Coordinate with lab members and faculty to plan and execute events promoting optics and photonics education. Previously participated as active member before being promoted to leadership position.
Financial relationship: No financial relationship; voluntary leadership position.

Nonlinear Fiber Optics Workshop May 2025 – July 2025
IKAA
Online
Nature and extent: Participated in intensive workshop on nonlinear fiber optics covering theory and simulations of light behavior in optical fibers from Maxwell's equations, nonlinear effects including self-phase modulation and four-wave mixing, using MATLAB and MEEP for fiber optic simulations.
Financial relationship: No financial relationship; participant only.

Summer Volunteer May 2023 – July 2023
Ek Pehal - IISER Kolkata
Kolkata, India
Nature and extent: Volunteered to teach underprivileged middle and high school kids as part of the Ek Pehal Foundation. Taught English, Maths and Physics for 6 hours per week.
Financial relationship: No financial relationship; volunteer position.

Physics through Computational Thinking February 2023 – April 2023
NPTEL - IISER Bhopal
Online

Nature and extent: Completed online course covering Wolfram Alpha and Fortran programming, Monte-Carlo simulations, statistical modeling, and numerical methods (Euler, Runge-Kutta) applied to physics problems.
Financial relationship: No financial relationship; course participant.

From Atoms to Materials: Predictive Theory and Simulations December 2023
edX - Purdue University
Online
Nature and extent: Completed online course applying classical mechanics, quantum mechanics and statistical mechanics to materials science. Conducted online DFT and molecular dynamics simulations using NanoHub to predict thermal and electronic properties of basic molecules.
Financial relationship: No financial relationship; course participant.

Winter School for Quantum Computing December 2022
IISER Kolkata
Kolkata, India
Nature and extent: Attended winter school covering basic quantum gates and quantum algorithms, decoherence challenges in quantum systems, and applications in medicine, finance, and material sciences.
Financial relationship: No financial relationship; participant only.

SKILLS AND COMPETENCIES

- **Experimental Techniques:** Dark-field polarization microscopy, Mueller matrix measurement, optical setup design, plasmonic characterization, spectroscopic analysis
- **Computational Physics:** Mie scattering theory implementation, electromagnetic simulation, optical property calculation, data analysis and visualization
- **Programming & Software:** MATLAB, Python, C, Inkscape, Bash scripting, Quantum ESPRESSO, WIEN2k, Origin, LaTeX, MySQL, HTML/CSS/JavaScript
- **Theoretical Knowledge:** Classical electrodynamics, quantum mechanics, nanophotonics, chiral optics, spin-orbit interactions, density functional theory
- **Quantum Technologies:** Quantum computing algorithms, quantum machine learning, Qiskit programming, quantum information processing

GRANTS AND AWARDS

- **Summer Research Fellowship 2024** - Indian Academy of Sciences (Completed at SVNIT Surat)
- Outstanding group assignment certificate for implementing Library Management System in C for course CS3101
- Event Organiser of SPIE Student Chapter, OPTICA community at IISER Kolkata
- **JEST Rank 165** - Joint Entrance Screening Test 2024
- **GRE Physics** - 850 total marks with 70 percentile
- **TOEFL** - Advanced English Fluency with 102/120 marks.
- **Gold Medal** - International English Olympiad
- Successfully cleared JEE Mains and Advanced examinations